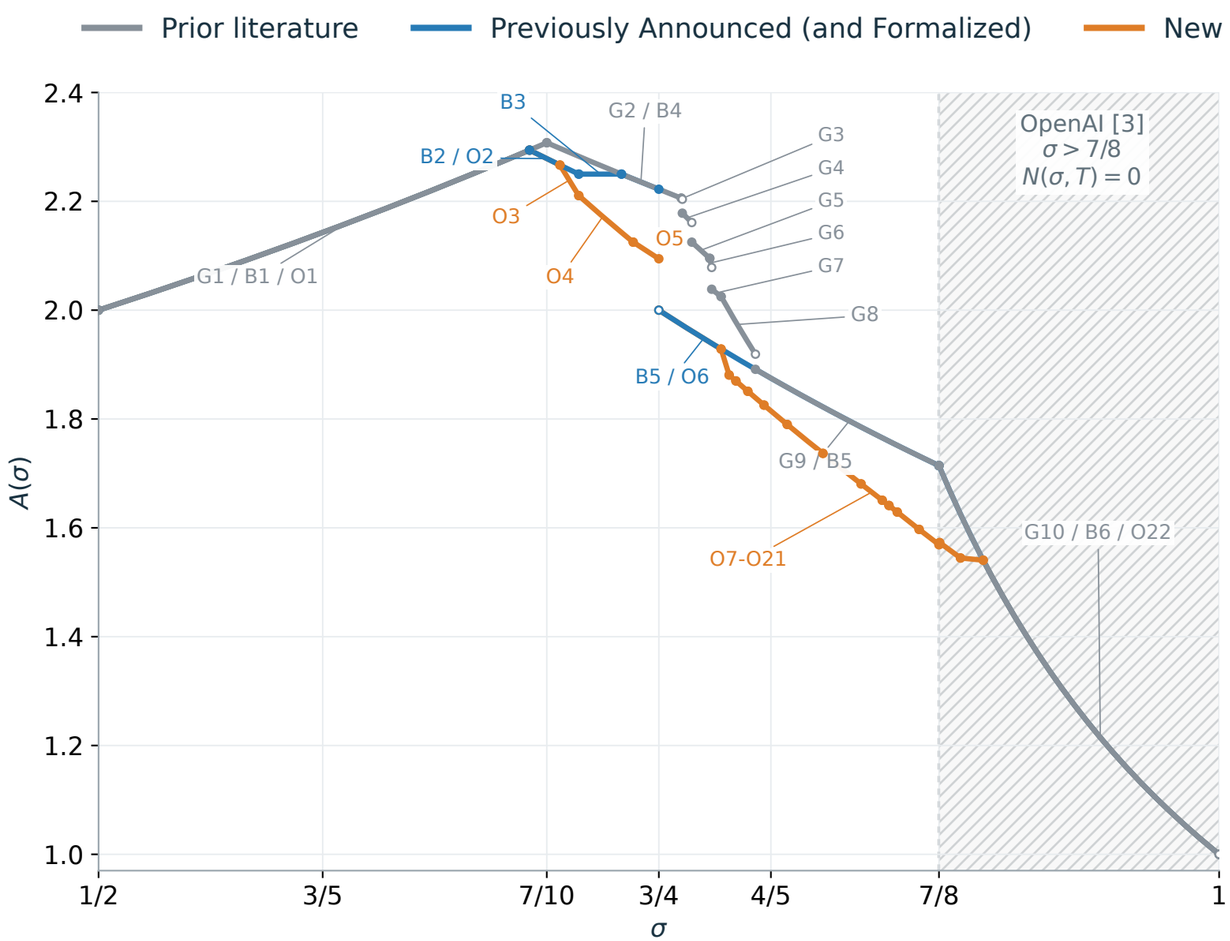
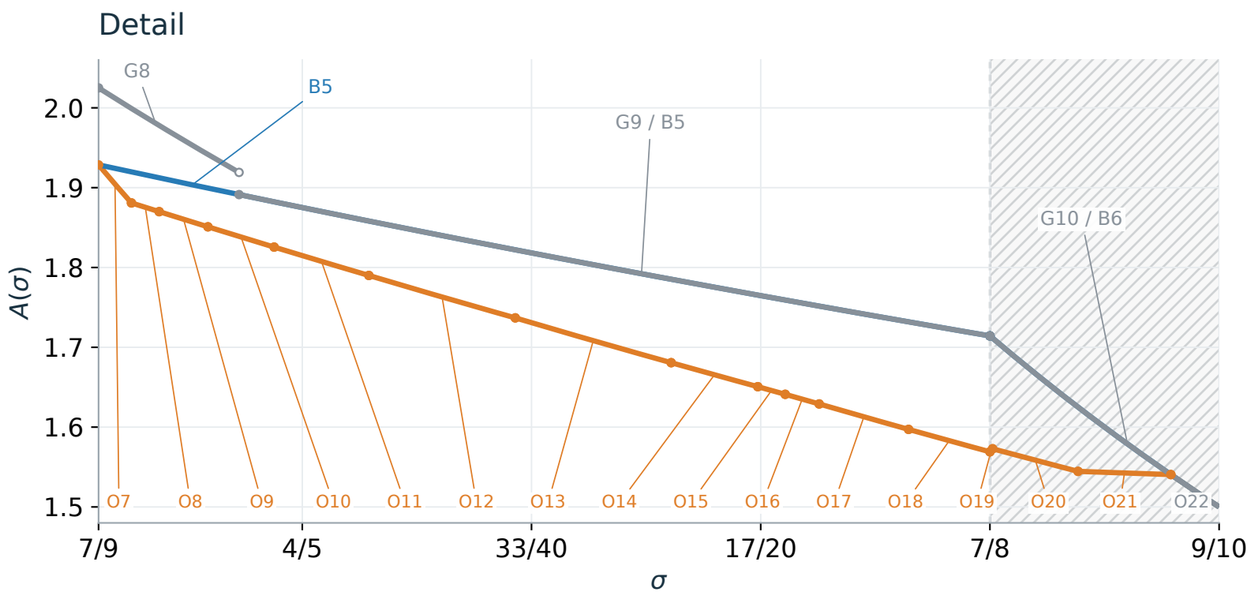
Cumulative zero-density power bounds

$$N(\sigma, T) \ll_{\sigma,\varepsilon} T^{(1-\sigma)A(\sigma)+\varepsilon}$$





Neutral hatching: OpenAI [3], σ > 7/8; boundary excluded.
OpenAI result; kernel-checked dependency; not a new contribution.

[1] PALOMAR-2026-10-11-000010 (v1)
[2] PALOMAR-2026-10-09-000005 (v1)

Estimates and their ranges

PRIOR LITERATURE

G1 [P1] $\frac{3}{2-\sigma}$ $[\frac{1}{2}, \frac{7}{10}]$
G2 [P2] $\frac{15}{5\sigma+3}$ $[\frac{7}{10}, \frac{19}{25}]$
G3 [P3] $\frac{9}{8\sigma-2}$ $[\frac{19}{25}, \frac{127}{167})$
G4 [P4] $\frac{15}{13\sigma-3}$ $[\frac{127}{167}, \frac{13}{17})$
G5 [P5] $\frac{6}{5\sigma-1}$ $[\frac{13}{17}, \frac{17}{22}]$
G6 [P6] $\frac{2}{9\sigma-6}$ $[\frac{17}{22}, \frac{41}{53})$
G7 [P7] $\frac{9}{7\sigma-1}$ $[\frac{41}{53}, \frac{7}{9}]$
G8 [P8] $\frac{9}{16\sigma-8}$ $[\frac{7}{9}, \frac{23}{29})$
G9 [P9] $\frac{3}{2\sigma}$ $[\frac{23}{29}, \frac{7}{8}]$
G10 [P10] $\frac{3}{10\sigma-7}$ $[\frac{7}{8}, 1)$

PREVIOUSLY ANNOUNCED (AND FORMALIZED)

B1 [P1] $\frac{3}{2-\sigma}$ $[\frac{1}{2}, \frac{9}{13}]$
B2 [P11] $\frac{18}{7\sigma+3}$ $[\frac{9}{13}, \frac{5}{7}]$
B3 [P12] $\frac{9}{4}$ $[\frac{5}{7}, \frac{11}{15}]$
B4 [P2] $\frac{15}{5\sigma+3}$ $[\frac{11}{15}, \frac{3}{4}]$
B5 [P13] $\frac{3}{2\sigma}$ $(\frac{3}{4}, \frac{7}{8}]$
B6 [P10] $\frac{3}{10\sigma-7}$ $[\frac{7}{8}, 1)$

NEW

O1 [P1] $\frac{3}{2-\sigma}$ $[\frac{1}{2}, \frac{9}{13}]$
O2 [P11] $\frac{18}{7\sigma+3}$ $[\frac{9}{13}, \frac{12}{17}]$
O3 [P14] $\frac{6}{8\sigma-3}$ $[\frac{12}{17}, \frac{5}{7}]$
O4 [P15] $\frac{12}{9\sigma-1}$ $[\frac{5}{7}, \frac{113}{153}]$
O5 [P16] $\frac{255}{153\sigma+7}$ $[\frac{113}{153}, \frac{3}{4}]$
O6 [P13] $\frac{3}{2\sigma}$ $(\frac{3}{4}, \frac{7}{9}]$
O7 [P17] $\frac{3}{11\sigma-7}$ $[\frac{7}{9}, \frac{29359}{37574}]$
O8 [P18] $\frac{4744}{4870\sigma-1283}$ $[\frac{29359}{37574}, \frac{1015}{1294}]$
O9 [P19] $\frac{6154}{6354\sigma-1693}$ $[\frac{1015}{1294}, \frac{353}{447}]$
O10 [P20] $\frac{3782}{3936\sigma-1065}$ $[\frac{353}{447}, \frac{467}{586}]$
O11 [P21] $\frac{2168}{2282\sigma-631}$ $[\frac{467}{586}, \frac{289}{358}]$
O12 [P22] $\frac{566}{606\sigma-173}$ $[\frac{289}{358}, \frac{163}{198}]$
O13 [P23] $\frac{31544}{35670\sigma-11203}$ $[\frac{163}{198}, \frac{7873}{9370}]$
O14 [P24] $\frac{1261710}{1447460\sigma-465509}$ $[\frac{7873}{9370}, \frac{2018}{2375}]$
O15 [P25] $\frac{3066}{3616\sigma-1215}$ $[\frac{2018}{2375}, \frac{1959705}{2298302}]$
O16 [P26] $\frac{597962}{729216\sigma-257415}$ $[\frac{1959705}{2298302}, \frac{6482825}{7570038}]$
O17 [P27] $\frac{16504356}{20732766\sigma-7623595}$ $[\frac{6482825}{7570038}, \frac{2905}{3354}]$
O18 [P28] $\frac{2172}{2742\sigma-1015}$ $[\frac{2905}{3354}, \frac{7}{8}]$
O19 [P29] $\frac{1086}{5537(1-\sigma)}$ $[\frac{7}{8}, \frac{24139}{27578}]$
O20 [P30] $\frac{24}{30\sigma-11}$ $[\frac{24139}{27578}, \frac{23}{26}]$
O21 [P31] $\frac{6}{\sigma+3}$ $[\frac{23}{26}, \frac{17}{19}]$
O22 [P10] $\frac{3}{10\sigma-7}$ $[\frac{17}{19}, 1)$

[3] OpenAI (30 September 2026), QRH; official scope: lean/docs/003.md

At $\sigma = 1$, $N(1, T) = 0$; the plotted A is a left limit.